

MSSQL

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/* =====
ECOMMERCE STORED FUNCTIONS LAB    Microsoft SQL Server
Version===== */
IF DB_ID('ecommerce_db') IS NULL
    CREATE DATABASE ecommerce_db;GO
USE ecommerce_db;GO

/* =====
DROP OLD TABLES===== */
DROP TABLE IF EXISTS order_items;
DROP TABLE IF EXISTS orders;
DROP TABLE IF EXISTS products;
DROP TABLE IF EXISTS customers;
GO

/* =====
CREATE TABLES===== */
CREATE TABLE customers (
    customer_id INT IDENTITY(1,1) PRIMARY KEY,
    customer_name VARCHAR(100),
    email VARCHAR(100),
    membership_level VARCHAR(30));
CREATE TABLE products (
    product_id INT IDENTITY(1,1) PRIMARY KEY,
    product_name VARCHAR(100),
    category VARCHAR(50),
    unit_price DECIMAL(10,2),
    stock_qty INT);
CREATE TABLE orders (
    order_id INT IDENTITY(1,1) PRIMARY KEY,
    customer_id INT,
    order_date DATE,
    order_status VARCHAR(30),
    CONSTRAINT FK_orders_customers
        FOREIGN KEY (customer_id) REFERENCES customers(customer_id));
CREATE TABLE order_items (
    order_item_id INT IDENTITY(1,1) PRIMARY KEY,
    order_id INT,
    product_id INT,
    quantity INT,
    unit_price DECIMAL(10,2),
    CONSTRAINT FK_order_items_orders
        FOREIGN KEY (order_id) REFERENCES orders(order_id),
    CONSTRAINT FK_order_items_products
        FOREIGN KEY (product_id) REFERENCES products(product_id));GO

/* =====
POPULATE SAMPLE DATA===== */
INSERT INTO customers (customer_name, email, membership_level)VALUES('Aung
Aung', 'aung@example.com', 'Gold'),('Su Su', 'su@example.com',
'Silver'),('Kyaw Kyaw', 'kyaw@example.com', 'Normal'),('Mya Mya',
'mya@example.com', 'Gold'),('Ko Ko', 'koko@example.com', 'Silver'),('Hla Hla',
'hla@example.com', 'Normal'),('Zaw Zaw', 'zaw@example.com', 'Gold'),('Ei Ei',
'eiei@example.com', 'Silver'),('Tun Tun', 'tun@example.com', 'Normal'),('May
May', 'may@example.com', 'Gold');

INSERT INTO products (product_name, category, unit_price,
stock_qty)VALUES('Laptop', 'Electronics', 1200000, 10),('Mouse', 'Electronics',
25000, 100),('Keyboard', 'Electronics', 45000, 50),('Office Chair',
'Furniture', 180000, 20),('Desk', 'Furniture', 250000, 15),('Monitor',
'Electronics', 350000, 25),('Printer', 'Electronics', 450000, 8),('USB
Keyboard', 'Electronics', 30000, 120),('Gaming Mouse', 'Electronics', 55000,
75),('Bookshelf', 'Furniture', 200000, 12),('Meeting Table', 'Furniture',
650000, 5),('Headphone', 'Electronics', 85000, 40),('Webcam', 'Electronics',

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95000, 18), ('Air Conditioner', 'Appliances', 1800000, 3), ('Projector',
'Electronics', 950000, 6);
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INSERT INTO orders (customer_id, order_date, order_status) VALUES(1, '2026-05-
01', 'Completed'), (2, '2026-05-02', 'Completed'), (3, '2026-05-03',
'Pending'), (4, '2026-05-04', 'Completed'), (5, '2026-05-05', 'Pending'), (6,
'2026-05-06', 'Cancelled'), (7, '2026-05-07', 'Completed'), (8, '2026-05-08',
'Completed'), (9, '2026-05-09', 'Pending'), (10, '2026-05-10', 'Completed');
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```
INSERT INTO order_items (order_id, product_id, quantity, unit_price) VALUES(1,
1, 1, 1200000), (1, 2, 2, 25000), (2, 4, 1, 180000), (2, 3, 1, 45000), (3, 5, 1,
250000), (4, 6, 1, 350000), (4, 7, 2, 450000),
(5, 8, 1, 30000), (5, 9, 2, 55000),
(6, 10, 1, 200000),
(7, 11, 1, 650000), (7, 12, 3, 85000),
(8, 13, 2, 95000),
(9, 14, 1, 1800000),
(10, 15, 1, 950000),
(11, 1, 1, 1200000), (11, 3, 2, 45000),
(12, 4, 1, 180000),
(13, 5, 1, 250000);
```

GOGO

```
/* =====
DROP OLD FUNCTIONS===== */
DROP FUNCTION IF EXISTS fn_order_subtotal; DROP FUNCTION IF EXISTS
fn_tax_amount; DROP FUNCTION IF EXISTS fn_membership_discount; DROP FUNCTION IF
EXISTS fn_order_final_total; DROP FUNCTION IF EXISTS fn_stock_status; DROP
FUNCTION IF EXISTS fn_customer_order_count; DROP FUNCTION IF EXISTS
fn_membership_label; DROP FUNCTION IF EXISTS fn_shipping_fee; DROP FUNCTION IF
EXISTS fn_order_grand_total;
GO
```

```
160 /* =====
161 FUNCTION 1: Calculate Order Subtotal
162 ===== */
163
164 CREATE FUNCTION fn_order_subtotal
165 (
166     @p_order_id INT
167 )
168 RETURNS DECIMAL(10,2)
169 AS
170 BEGIN
171     DECLARE @v_subtotal DECIMAL(10,2);
172
173     SELECT @v_subtotal = SUM(quantity * unit_price)
174     FROM order_items
175     WHERE order_id = @p_order_id;
176
177     RETURN ISNULL(@v_subtotal, 0);
178 END;
179 GO
180
181 SELECT dbo.fn_order_subtotal(1) AS order_subtotal;
182 GO
183
```

99 % No issues found Ln: 10, Ch: 18 SPC CRLF UTF-8

order_subtotal
1 0.00

```
184 /* =====
185 FUNCTION 2: Calculate Tax Amount
186 ===== */
187
188 CREATE FUNCTION fn_tax_amount
189 (
190     @p_amount DECIMAL(10,2)
191 )
192 RETURNS DECIMAL(10,2)
193 AS
194 BEGIN
195     RETURN @p_amount * 0.05;
196 END;
197 GO
198
199 SELECT dbo.fn_tax_amount(100000) AS tax_amount;
200 GO
201
```

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tax_amount
1 5000.00

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202  /* =====
203  FUNCTION 3: Membership Discount
204  ===== */
205
206  CREATE FUNCTION fn_membership_discount
207  (
208      @p_amount DECIMAL(10,2),
209      @p_membership_level VARCHAR(30)
210  )
211  RETURNS DECIMAL(10,2)
212  AS
213  BEGIN
214      DECLARE @v_discount DECIMAL(10,2);
215
216      IF @p_membership_level = 'Gold'
217          SET @v_discount = @p_amount * 0.10;
218      ELSE IF @p_membership_level = 'Silver'
219          SET @v_discount = @p_amount * 0.05;
220      ELSE
221          SET @v_discount = 0;
222
223      RETURN @v_discount;
224  END;
225  GO
226
227  SELECT dbo.fn_membership_discount(100000, 'Gold') AS discount_amount;
228  GO

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99 % No issues found

Ln: 10, Ch: 18 SPC CRLF UTF-8

Results Messages

discount_amount
1 10000.00

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230  /* =====
231  FUNCTION 4: Final Order Total
232  ===== */
233
234  CREATE FUNCTION fn_order_final_total
235  (
236      @p_order_id INT
237  )
238  RETURNS DECIMAL(10,2)
239  AS
240  BEGIN
241      DECLARE @v_subtotal DECIMAL(10,2);
242      DECLARE @v_tax DECIMAL(10,2);
243      DECLARE @v_discount DECIMAL(10,2);
244      DECLARE @v_membership VARCHAR(30);
245      DECLARE @v_final_total DECIMAL(10,2);
246
247      SELECT @v_membership = c.membership_level
248      FROM orders o
249      JOIN customers c ON o.customer_id = c.customer_id
250      WHERE o.order_id = @p_order_id;
251
252      SET @v_subtotal = dbo.fn_order_subtotal(@p_order_id);
253      SET @v_tax = dbo.fn_tax_amount(@v_subtotal);
254      SET @v_discount = dbo.fn_membership_discount(@v_subtotal, @v_membership);
255
256      SET @v_final_total = @v_subtotal + @v_tax - @v_discount;
257
258      RETURN ISNULL(@v_final_total, 0);
259  END;
260  GO
261
262  SELECT dbo.fn_order_final_total(1) AS final_order_total;
263  GO

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Results Messages

final_order_total
1 0.00

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265  /* =====
266  FUNCTION 5: Stock Status
267  ===== */
268
269  CREATE FUNCTION fn_stock_status
270  (
271      @p_stock_qty INT
272  )
273  RETURNS VARCHAR(30)
274  AS
275  BEGIN
276      RETURN
277      CASE
278          WHEN @p_stock_qty = 0 THEN 'Out of Stock'
279          WHEN @p_stock_qty < 10 THEN 'Low Stock'
280          ELSE 'Available'
281      END;
282  END;
283  GO
284
285  SELECT
286      product_name,
287      stock_qty,
288      dbo.fn_stock_status(stock_qty) AS stock_status
289  FROM products;
290  GO
291

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Results Messages

product_name	stock_qty	stock_status
4 Office Chair	20	Available
5 Desk	15	Available
6 Monitor	25	Available
7 Printer	8	Low Stock
8 USB Keyboard	120	Available
9 Gaming Mou...	75	Available
10 Bookshelf	12	Available
11 Meeting Table	5	Low Stock

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292  /* =====
293  FUNCTION 6: Customer Order Count
294  ===== */
295
296  CREATE FUNCTION fn_customer_order_count
297  (
298      @p_customer_id INT
299  )
300  RETURNS INT
301  AS
302  BEGIN
303      DECLARE @v_total_orders INT;
304
305      SELECT @v_total_orders = COUNT(*)
306      FROM orders
307      WHERE customer_id = @p_customer_id;
308
309      RETURN @v_total_orders;
310  END;
311  GO
312
313  SELECT
314      customer_name,
315      dbo.fn_customer_order_count(customer_id) AS total_orders
316  FROM customers;
317  GO
318
```

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customer_name	total_orders
Aung Aung	1
Su Su	1
Kyaw Kyaw	1
Mya Mya	1
Ko Ko	1
Hla Hla	1
Zaw Zaw	1
Ei Ei	1

```
319  /* =====
320  FUNCTION 7: Membership Label
321  ===== */
322
323  CREATE FUNCTION fn_membership_label
324  (
325      @p_membership_level VARCHAR(30)
326  )
327  RETURNS VARCHAR(100)
328  AS
329  BEGIN
330      RETURN
331      CASE
332          WHEN @p_membership_level = 'Gold' THEN 'Gold Member - 10% Discount'
333          WHEN @p_membership_level = 'Silver' THEN 'Silver Member - 5% Discount'
334          ELSE 'Normal Member - No Discount'
335      END;
336  END;
337  GO
338
339  SELECT
340      customer_name,
341      membership_level,
342      dbo.fn_membership_label(membership_level) AS membership_description
343  FROM customers;
344  GO
345
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customer_name	membership_level	membership_description
Kyaw Kyaw	Normal	Normal Member - No Disco...
Mya Mya	Gold	Gold Member - 10% Discount
Ko Ko	Silver	Silver Member - 5% Discount
Hla Hla	Normal	Normal Member - No Disco...
Zaw Zaw	Gold	Gold Member - 10% Discount
Ei Ei	Silver	Silver Member - 5% Discount
Tun Tun	Normal	Normal Member - No Disco...
May May	Gold	Gold Member - 10% Discount

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346  /* =====
347  FUNCTION 8: Shipping Fee
348  ===== */
349
350  CREATE FUNCTION fn_shipping_fee
351  (
352      @p_amount DECIMAL(10,2)
353  )
354  RETURNS DECIMAL(10,2)
355  AS
356  BEGIN
357      IF @p_amount >= 500000
358          RETURN 0;
359
360      RETURN 5000;
361  END;
362  GO
363
364  SELECT dbo.fn_shipping_fee(300000) AS shipping_fee;
365  GO
366
```

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shipping_fee
5000.00

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367  /* =====
368  FUNCTION 9: Grand Total
369  ===== */
370
371  CREATE FUNCTION fn_order_grand_total
372  (
373    @p_order_id INT
374  )
375  RETURNS DECIMAL(10,2)
376  AS
377  BEGIN
378    DECLARE @v_final_total DECIMAL(10,2);
379    DECLARE @v_shipping_fee DECIMAL(10,2);
380
381    SET @v_final_total = dbo.fn_order_final_total(@p_order_id);
382    SET @v_shipping_fee = dbo.fn_shipping_fee(@v_final_total);
383
384    RETURN @v_final_total + @v_shipping_fee;
385  END;
386  GO
387
388  SELECT dbo.fn_order_grand_total(1) AS grand_total;
389  GO
390
391  /* =====

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Results Messages

	grand_total
1	5000.00

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391  /* =====
392  PRACTICAL REPORT QUERY
393  ===== */
394
395  SELECT
396    o.order_id,
397    c.customer_name,
398    c.membership_level,
399    dbo.fn_order_subtotal(o.order_id) AS subtotal,
400    dbo.fn_tax_amount(dbo.fn_order_subtotal(o.order_id)) AS tax_amount,
401    dbo.fn_membership_discount(
402      dbo.fn_order_subtotal(o.order_id),
403      c.membership_level
404    ) AS discount_amount,
405    dbo.fn_order_final_total(o.order_id) AS final_total,
406    dbo.fn_shipping_fee(dbo.fn_order_final_total(o.order_id)) AS shipping_fee,
407    dbo.fn_order_grand_total(o.order_id) AS grand_total,
408    o.order_status
409  FROM orders o
410  JOIN customers c ON o.customer_id = c.customer_id;
411  GO
412

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Results Messages

order_id	customer_name	membership_level	subtotal	tax_amount	discount_amount	final_total	shipping_fee	grand_total	order_status
1	Aung Aung	Gold	0.00	0.00	0.00	0.00	5000.00	5000.00	Completed
2	Su Su	Silver	0.00	0.00	0.00	0.00	5000.00	5000.00	Completed
3	Kyaw Kyaw	Normal	0.00	0.00	0.00	0.00	5000.00	5000.00	Pending
4	Mya Mya	Gold	0.00	0.00	0.00	0.00	5000.00	5000.00	Completed
5	Ko Ko	Silver	0.00	0.00	0.00	0.00	5000.00	5000.00	Pending
6	Hla Hla	Normal	0.00	0.00	0.00	0.00	5000.00	5000.00	Cancelled
7	Zaw Zaw	Gold	0.00	0.00	0.00	0.00	5000.00	5000.00	Completed
8	Ei Ei	Silver	0.00	0.00	0.00	0.00	5000.00	5000.00	Completed

```

413  /* =====
414  SHOW ALL USER FUNCTIONS
415  ===== */
416
417  SELECT
418    name AS function_name,
419    create_date,
420    modify_date
421  FROM sys.objects
422  WHERE type IN ('FN', 'IF', 'TF')
423  ORDER BY name;
424  GO

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Results Messages

function_name	create_date	modify_date
fn_membership_discount	2026-05-17 17:25:21.617	2026-05-17 17:25:21.617
fn_membership_label	2026-05-17 17:25:21.897	2026-05-17 17:25:21.897
fn_order_final_total	2026-05-17 17:25:21.663	2026-05-17 17:25:21.663
fn_order_grand_total	2026-05-17 17:25:21.963	2026-05-17 17:25:21.963
fn_order_subtotal	2026-05-17 17:25:21.517	2026-05-17 17:25:21.517
fn_shipping_fee	2026-05-17 17:25:21.933	2026-05-17 17:25:21.933
fn_stock_status	2026-05-17 17:25:21.730	2026-05-17 17:25:21.730
fn_tax_amount	2026-05-17 17:25:21.567	2026-05-17 17:25:21.567